

WILLIAM MALLARD

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Computational biologist and systems engineer building scalable analytical systems for multi-omics integration, predictive modeling, and ML on biological data. Two decades spanning HPC, software, and the lab bench.

EDUCATION

Harvard University PhD, Biochemistry	Cambridge, MA 8/2016 – 5/2025
University of California, Berkeley BSc, Engineering Physics; <i>minors in Computer Science and Mathematics</i>	Berkeley, CA 1/2002 – 5/2007

SKILLS

Languages	Python, C, C++, CUDA, Assembly, Java, Matlab, Bash
Frameworks	NumPy, SciPy, Pandas, PyTorch, scikit-learn, scikit-image
Systems	Linux, GPUs, HPC/SLURM, Git
Domains	Computational Biology, Transcriptomics, Proteomics, Structural Biology, Predictive Modeling, Machine Learning, Computer Vision

PROJECTS

- bacterial phospho-site predictor** (2026) — Trained an MLP on ESM-2 embeddings to predict phospho-sites in bacteria, sweeping hyperparameters to find a minimal architecture. Discovered that a published ten-feature ensemble’s benchmark was inflated by train-on-test contamination; showed that my simpler model matched its performance when both were run on a properly split version of their dataset.
- lipid-droplet proximity proteomics** (2024) — Devised a candidate-selection method for proteomics timecourse data, starting from ~5,700 proteins across a 6-timepoint TurboID/TMT mass-spec dataset. Represented each protein as an “enrichment trajectory” in volcano space, and then ROC-tuned a hyperbolic threshold against a known reference proteome to select candidates. Prioritized 20 candidate lipid-droplet surface proteins, 17 of which validated via live-cell imaging.
- discourse-analysis** (2026) — Built a search engine over a Twitter corpus (~500k posts) unifying full-text, fuzzy, and semantic search. Runs image OCR (Tesseract), machine translation (NLLB-200), and embedding (Qwen) on-device at zero cloud cost. Retrieves posts and reconstructs their context (threads, replies, quotes). Exposes an agentic tool interface (MCP) tailored to how an LLM explores a corpus.
- gpu-xengine** (2026) — Ported a matrix multiplication engine for real-time signal cross-correlation from FPGA to CUDA. Optimized memory access patterns (coalescing, pinning) for a $1.5\times$ throughput gain.

EXPERIENCE

Harvard University <i>Graduate Student Researcher</i> , Molecular and Cellular Biology	Cambridge, MA 8/2016 – 12/2025
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Led a multi-year research project from conception through publication, discovering a novel regulatory mechanism in bacterial cell division (PLOS One, 2025).

- **Structural prediction and validation:** Sampled 5,000 AlphaFold-Multimer conformations of an IDR bound to a globular domain, and built a contact-map clustering pipeline that identified candidate binding modes and interface residues. Confirmed the predicted interface experimentally via NMR and ITC; built pipelines for quantitative 2D-NMR analysis of a phospho-mutant series to resolve the structural mechanism.
- **Image analysis and microscopy:** Built computer vision and computational geometry pipelines; hand-wrote core algorithms in vectorized NumPy; profiled to eliminate bottlenecks; parallelized processing on a SLURM cluster. Used these to analyze live-cell fluorescence, TIRF, and confocal imaging experiments that I also designed and ran.

- **Comparative genomics:** Scanned the *B. subtilis* proteome to catalog its Ser/Thr kinases, identifying a previously-unannotated fourth kinase. Combined active-site motif analysis, structural comparison, and phylogenetics to narrow the candidate FtsZ kinases to one, subsequently confirmed by knockout.
- **Scientific writing:** Independently managed the entire writing, revision, and peer review process as corresponding author (6/2025–12/2025).

Tools: Python, NumPy, Pandas, SciPy, scikit-image, Fiji, Jython, AlphaFold, SLURM, Bash, NMRPipe, Poky

Broad Institute of MIT and Harvard

Cambridge, MA

Computational Biologist, Rinn Lab

2/2013 – 7/2016

Contributed genomics analyses to 13 publications across stem-cell biology, epigenetics, hematopoiesis, and immunogenetics, working with bench scientists across many labs.

- Designed integrative analyses of RNA-seq, ChIP-seq, ATAC-seq, and bisulfite-seq data — tailored to each collaborator's question (e.g., differential expression, differential binding, peak-to-gene mapping).
- Analyzed data end-to-end: from demultiplexing and QC, to alignment and quantification, to genome arithmetic and bespoke downstream analyses; produced tables and figure panels; handled SRA/GEO submissions.
- Processed ~10TB across 20+ projects; wrote a Python C extension for LSF job management to keep large analyses running at scale.

Tools: Python, NumPy, Pandas, Bowtie, TopHat, Cufflinks, bedtools, Unix tools (grep, sed, awk), C, LSF

Broad Institute of MIT and Harvard

Cambridge, MA

Software Engineer, The Cancer Genome Atlas

4/2012 – 2/2013

Maintained and extended the GDAC Firehose pipeline, a production genomics platform processing tens of thousands of samples across dozens of cancer types.

- Triaged pipeline failures across automated analysis runs, diagnosing and resolving performance bottlenecks in data ingestion and processing stages.
- Rewrote memory-bound R scripts as streaming Python, scaling them to arbitrarily many samples.
- Traced a production failure to a Python standard library limitation; contributed fixes to CPython 3.4 that enabled large 64-bit archives (shutil, #17189) and made them the default (zipfile, #17201).

Tools: Python, NumPy, R, Bash

Space Sciences Laboratory

Berkeley, CA

Systems Engineer, Townes Lab

9/2009 – 8/2011

Built a real-time FX correlator for the Infrared Spatial Interferometer at Mount Wilson Observatory.

- Designed a 138 Gbps signal processing system across 3 synchronized Xilinx Virtex-5 FPGAs, channelizing 2.88 GHz of bandwidth into 64 spectral channels with real-time cross-correlation
- Achieved a 1.8× clock speedup (200 → 360 MHz) over existing CASPER designs by mapping core FFT arithmetic onto DSP48Es and developing a BRAM/DSP floorplanning methodology
- Contributed 50+ optimized DSP commits upstream to CASPER's open-source FPGA library; documented floorplanning methodology in CASPER Memo #42
- Built an end-to-end data acquisition pipeline, from FPGA fabric to PowerPC firmware to Python control software and networked storage
- Instrument resolved individual molecular signatures in Betelgeuse's atmosphere; results published in Wishnow et al. 2010 (SPIE); code and documentation available on GitHub.

Tools: Simulink, MATLAB, C, Python

Center for Astronomy Signal Processing and Electronics Research

Berkeley, CA

Systems Engineer, Werthimer Lab

4/2008 – 9/2009

Contributed to CASPER, an open-source FPGA toolkit for radio astronomy signal processing.

- Developed a 10 Gbps UDP receiver with a synchronized ring buffer; core architecture adopted into PySPEAD, the data transport layer for the Square Kilometre Array

- Refactored and extended Simulink blocks and mask scripts in the CASPER FPGA library
- Built enclosure and RF signal routing for a spectrometer deployed at Arecibo Observatory

Tools: Simulink, MATLAB, C, Python

MusicianLink

Software Engineer

San Jose, CA

6/2007 – 3/2008

Designed the binary protocol and connection manager for a real-time networked music device.

UC Berkeley Clustered Computing

Unix Systems Administrator

Berkeley, CA

1/2006 – 8/2006

Administered Unix systems, server hardware, and job schedulers for scientific HPC clusters.

Massachusetts Institute of Technology

Research Assistant, Chuang Lab

Cambridge, MA

6/2005 – 8/2005

Built a full-stack instrument to shuttle ions in a planar Paul trap: designed an instruction set, wrote an interpreter in microcontroller assembly, and fabricated the hardware (Phys. Rev. A, 2006).

University of California, Berkeley

Research Assistant, Holzapfel Lab

Berkeley, CA

6/2004 – 5/2005

Wrote a C++ closed-loop control system to sequence the cooldown of a helium dilution fridge and hold a millimeter-wave telescope detector at 250 mK.

SELECTED PUBLICATIONS

Mallard WJ, Pham VV. "FtsZ phosphorylation modulates tail-core binding to tune cell division in *Bacillus subtilis*." *PLoS One* (2025).

Choudhuri A, Trompouki E, Abraham BJ, Colli LM, Kock KH, **Mallard W**, . . . , Rinn J, Albert PS, Bulyk ML, Chanock SJ, Young RA, Zon LI. "Common variants in signaling transcription-factor-binding sites drive phenotypic variability in red blood cell traits." *Nature Genetics* (2020).

Kaewsapsak P, Shechner DM, **Mallard W**, Rinn JL, Ting AY. "Live-cell mapping of organelle-associated RNAs via proximity biotinylation combined with protein-RNA crosslinking." *eLife* (2017).

Choi J, Lee S, **Mallard W**, Clement K, Tagliazucchi GM, Lim H, Choi IY, Ferrari F, Tsankov A, Pop R, Lee G, Rinn JL, Meissner A, Park P, Hochedlinger K. "A comparison of genetically matched cell lines reveals the equivalence of human iPSCs and ESCs." *Nature Biotechnology* (2015).

Cancer Genome Atlas Research Network. "Comprehensive molecular profiling of lung adenocarcinoma." *Nature* (2014).

Wishnow EH, **Mallard W**, Ravi V, Lockwood S, Fitelson W, Wertheimer D, Townes CH. "Mid-infrared interferometry with high spectral resolution." *Proc SPIE* (2010).

Pearson C, Leibrandt D, Bakr W, **Mallard W**, Brown K, Chuang I. "Experimental Investigation of Planar Ion Traps." *Physical Review A* (2006).

Full list available on Google Scholar.